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ICAM

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TYPE

Pre File

# The PANTONE process

*Exhaust gases in the intake*

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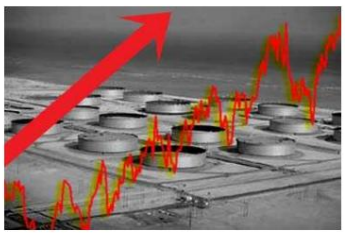
# I - Introduction



Since the acceleration of the development of emerging countries such as China or India, many studies predict the depletion of oil resources for the years to come.
Estimates vary greatly depending on the laboratory, but it can be said that oil will become an increasingly rare and increasingly expensive commodity.

Many teams of engine engineers are therefore looking into other means of producing mechanical energy such as hydrogen engines or induction systems.

However, the vehicle fleet cannot be changed in time. Even assuming that a revolutionary process is discovered soon, it would take about twenty years to change the global car fleet.

Certain processes have therefore been developed to increase the yields of heat engines at a lower cost. Of these, we chose PMC Pantone for several reasons. First of all, it is a process that is really easy to set up because it only involves, roughly, a few pipes and a can. We can therefore imagine quickly updating existing engines for a derisory cost. Then, the PMC Pantone has a huge ecological advantage. Indeed, by reusing the exhaust gases and reinjecting them into the intake, we will reduce atmospheric pollution and make it almost zero for certain settings. Finally, the PMC Pantone will make it possible to greatly reduce fuel consumption since on the best models we manage to operate engines with a mixture made up of less than 20% hydrocarbons and more than 80% water.

It is the principle of reusing exhaust gases in the intake which constitutes the duality studied in this dossier. Indeed, it may at first appear eccentric to want to inject used gases into the intake, whereas one usually works to send only the purest possible air.

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## II - Our study

We will therefore develop our presentation in three parts:

The first part is a general presentation of PMC Pantone. We will first present the context of its discovery as well as the efforts made to make this process known. We will then see the presentation of the mechanism, its composition and a brief overview of its operation.

The second part consists of a more in-depth study of the reaction that takes place in the PMC, and in particular in the “plasma reactor with endothermic reaction” part. We will then proceed to a thermodynamic study allowing us to determine the advantages of PMC in terms of fuel savings.

The third part will consist of a study experimental PMC, where we will try to design and then implement a PMC on a mower engine type engine. For this we will be content to use the plans put into free circulation by Pantone, the spec program. not allowing us to complete our studies at 100% and therefore we will not have the possibility of improving or adapting the system.

## III – Presentation and explanation of the process

The Multi-Fuel Processor is a new internationally patented technology (US005794601A1) whose inventor is Paul Pantone. This is a system that recovers the exhaust gases and reinjects them into the intake after subjecting them to a few chemical reactions.

Based on the principle of reusing exhaust gases instead of rejecting them, PMC Pantone reduces pollution by exhaust gases by nearly 85% compared to a traditional engine.

This system allows most conventional 4-stroke engines to operate with a water/ hydrocarbon mixture. It is capable of using all types of fuels (gasoline, diesel, kerosene, used oils and other hydrocarbon derivatives, etc.) thanks to its endothermic reaction plasma reactor. Tests carried out by numerous manufacturers and experimenters have already shown that it is even possible to operate an engine equipped with this device with a mixture of hydrocarbons and water whose proportions of water can reach 80%.

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The device is made up of three main parts: • The Intake/Exhaust connection, • The endothermic reaction plasma reactor (containing the magnetic bar and the pyrolysis chamber), • The bubbler.

The conventional carburettor and the muffler (and its catalytic system) are no longer necessary after installation of the device.

The bubbler is a tank containing a mixture of water and hydrocarbons. Proportions will generally be determined by experience, but the many tests carried out by experimenters show that a mixture containing only 20% hydrocarbons is perfectly usable….

Endothermic reaction plasma reactor may seem like a big word. However, we will see that, even if the operation is quite difficult to study, the principle nevertheless remains quite simple.

The reactor is made up of two coaxial steel cylinders: • The inner cylinder (threaded at each end), called the pyrolysis chamber, which contains a steel rod. The rod is held in the center of the pyrolysis chamber by 3 small studs welded to each of its ends. • The outer cylinder (threaded at each end) is a steel tube.

The two cylinders are placed coaxially thanks to two T reducers placed at each end.

Its operation is based on two principles. On one side, we will heat a tube inside by the exhaust gases which will pass through the outer tube (this is the same principle as observed in a distillation column cooler, but with the aim of heating, etc.). On the other hand, on the return circuit, a reduction in the diameter of the tube will cause a venturi effect which will heat the gas.

These two combined effects will bring the gas to a very high temperature while providing it with a sufficient quantity of energy to obtain thermochemical dissociation of the water-fuel mixture.

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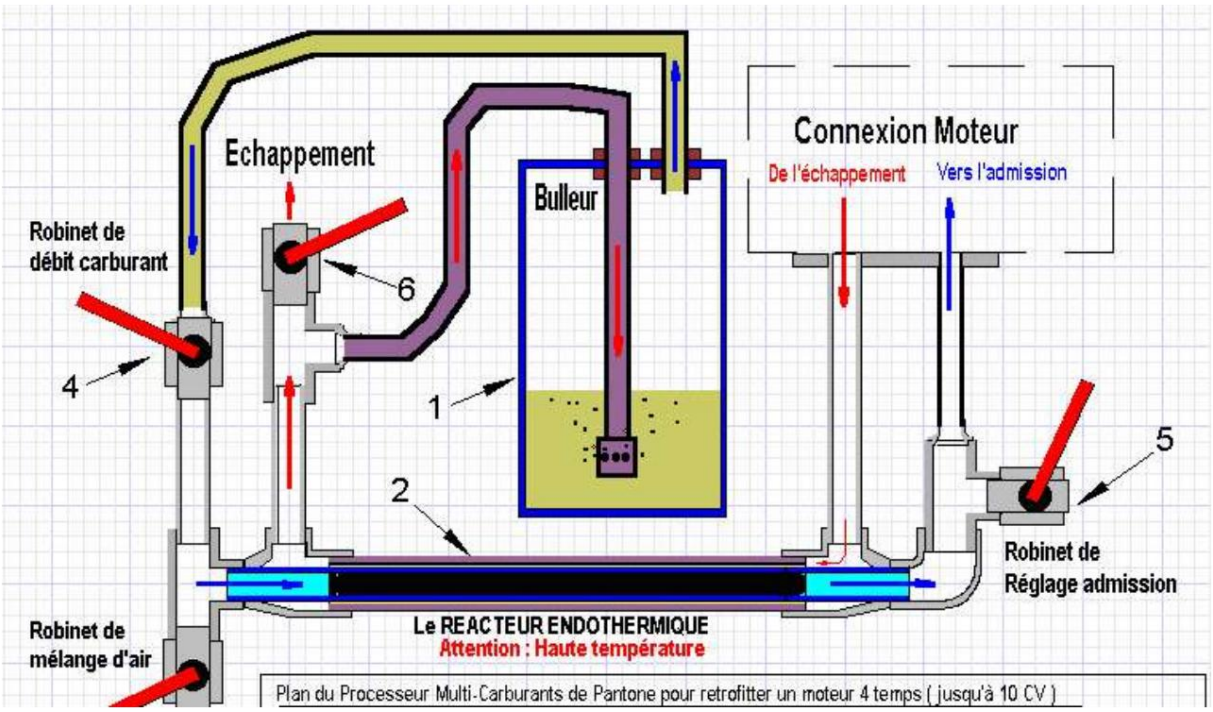


The flow of hot gases from the engine exhaust circulates through the outer part of the reactor with high kinetic energy (due to the small passage space: venturi effect), this contributes to bringing the steel rod to a very high temperature ( serving as a heat accumulator) contained in the pyrolysis chamber. The gases pass through the reactor and then enter the bubbler containing the water/fuel mixture. The vapors of the mixture are drawn in strongly by the vacuum created by the intake and pushed by the pressure coming from the exhaust. The kinetic energy of the vapors is increased considerably (in the light blue part) by the reduction of the diameter in the pyrolysis chamber (Venturi effect).

The combined effect of the high temperature and this increased kinetic energy causes the thermochemical decomposition of the water/fuel mixture.

The reactor therefore makes it possible to create a fuel with high energy efficiency through the decomposition of the water contained in the mixture. We therefore have a gas composed of O2 (present in large quantities in the exhaust gases) as well as a certain proportion of hydrogen.

The endothermic reactor also undergoes magnetization following its operation, this phenomenon is still little explained and we therefore prefer to neglect it for the moment, pending confirmation from Mr Fauchille as to whether this magnetization has any effect on the dissociation of water molecules.

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**Two disadvantages: starting and acceleration.**

There is a small difficulty during startup. Indeed, it takes approximately one minute for the endothermic reactor to reach its operating temperature (about 800°C). There is therefore a period of "heating up" during which the engine does not reach its optimum performance and where an injection of pure fuel may be necessary in the intake. This injection also appears to be essential for the first few seconds until the circulation of the gases takes place.

It will also be noted that, the mechanism removing the conventional carburettor, there is no longer a throttle lever. The engine speed will indeed be adjusted using the taps located on the device.

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## V - Echéancier

- Develop contact with Mr Fauchille to have explanations on thermochemical dissociation and its use in PMC Pantone.

- Contact the Energy department for more information on fluid mechanics, the venturi effect and gas circulation.

- Succeed in obtaining a visit to the ARRAS farmer who currently charges visits to show his tractor equipped with the PMC.

- Get in touch with one or more French experimenters to share their opinions and get help in carrying out our own project

- Realization of a model to explain the principle of the endothermic plasma reactor.

- In collaboration with an ICAM 1 motivated by the process, realization of a prototype on a mower engine type engine, then possibly on a Diesel 1L9. Hypothetical adaptation on a vehicle running in the current year (after checking the legal texts).